



Full Paper

# Deep Learning Models for the Identification of Biomarker, Drug-Target Interaction, and Pharmacokinetics in Pancreatic Ductal Adenocarcinoma

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# Deep Learning Models for the Identification of Biomarker, Drug-Target Interaction, and Pharmacokinetics in Pancreatic Ductal Adenocarcinoma

**Abstract**— Pancreatic ductal adenocarcinoma (PDAC) develops in the ductal cells of the pancreas, which are responsible for carrying digestive enzymes. It is considered highly aggressive due to late diagnosis, poor survival rate, early metastasis, mostly unresectable, and responds poorly to current treatments. In this study, we attempted to identify the targets of PDAC, drug-target interaction (DTI), and pharmacokinetics of the identified drugs using deep learning models. Initially, we considered three microarray gene expression datasets publicly available in the NCBI GEO and pre-processed using a standard pipeline. We identified the differentially expressed genes between normal and PDAC using deep learning models such as long short-term memory (LSTM), recurrent neural network (RNN), and one-dimensional convolutional neural network (1D-CNN). Further, the drug library was prepared for identified targets using the DepMap database. The embeddings of targets were generated using ProtVec, ProtBert, evolutionary scale modeling (ESM), and drugs were made using Mol2Vec, graph isomorphism network (GIN), and Unimol. We performed the DTI analysis and pharmacokinetics of the identified drugs using the same deep learning models. Our study identified 22 biomarker genes with a balanced accuracy of 98.89% using RNN. We found 106 drugs based on DTI analysis with a high  $R^2$  of 0.86 using Mol2Vec+GIN+Unimol+ProtVec+ProtBert+ESM and RNN. Our models were able to predict the pharmacokinetics properties such as absorption, distribution, metabolism, excretion, and toxicity with a mean absolute error of 1.62 (GIN+Mol2Vec+RNN), Spearman correlation (SC) of 0.56 (Mol2Vec+LSTM), area under the curve of 62.5% (Mol2Vec+Unimol+1D-CNN), SC of 0.32 (Unimol+LSTM), and 0.36 (GIN+Mol2Vec+1D-CNN), respectively. Our study demonstrates the potential of deep learning models as an effective and scalable solution for prognostic marker identification, disease-specific drug-target interaction prediction, and pharmacokinetics prediction, with promising implications for drug repurposing in PDAC.

**Index Terms**— Pancreatic ductal adenocarcinoma, Biomarker, Drug target interaction, Pharmacokinetics, Deep learning

## I. INTRODUCTION

**P**ancreatic ductal adenocarcinoma (PDAC) is a highly aggressive and third leading cause of cancer deaths worldwide[1]. The number of cases goes up by 1.1% per year and PDAC makes up nearly 19% of all new gastrointestinal cancers[2]. The current diagnostic approaches for PDAC primarily involve imaging modalities such as

magnetic resonance imaging, computed tomography, endoscopic retrograde cholangiopancreatography, endoscopic ultrasonography, and laparoscopic evaluation[3]. In addition, laboratory assessments focus on biomarkers including CA 19-9, CA-125, and carcinoembryonic antigen[4]. On the other hand, these biomarkers of PDAC detection are not limited to only PDAC. It can be elevated in other cancer conditions and disease environments[5, 6]. There is an urgent need to design and develop new personalized diagnosis and treatment models for PDAC[6].

The transcriptomic biomarkers play a very crucial role in diagnosis and targeted therapy for cancers. It can help in recognizing genes involved in PDAC progression, assist in identifying high-risk patients, and optimize personalized therapeutic strategies[7]. The gene expression data can be generated through microarray and RNA-seq techniques. The RNA-seq has limitations such as difficulty in counting small and overlapping transcripts, and different transcripts from the same gene[8]. On the other hand, microarray data can capture gene expression data for a large panel of genes and give insight into the biological information[8]. In this study, we considered the gene expression data obtained through microarray approach to identify biomarkers for PDAC.

The differentially expressed genes (DEGs) or targets are genes that show significant differences in expression levels between different biological conditions, such as healthy and PDAC[9]. The DEGs are identified using fold change and statistical analysis using different packages in R[10]. However, fold-change thresholds do not account for variability or ensure reproducibility, leading to the widespread use of statistical tests such as the t-test and the Wilcoxon test. These methods, however, were later found to produce high false discovery rates in small sample sizes and show only a weak relationship with fold-change[11]. Several tools, such as limma, DESeq2, edgeR, baySeq, and GEO2R, were also reported for the identification of DEGs [9]. The disadvantages of using these tools are as it depends on the selection process based on the specific needs of the study and factors like sample size, biological variability, and the underlying assumptions of the data distribution[9]. Studies have also used supervised machine learning (ML) methods to select the genes (DEGs) responsible for the disease[12]. ML

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models such as extreme gradient boost, random forest, Naive Bayes, and support vector machines[13]. ML models may fail while patient grouping is based on a non-linear relationship with a feature (i.e., gene). These features may be hidden to DEG analysis, leading to a failure of the subsequent downstream analysis and validation and, overall, an incomplete biological characterization [14]. Further, deep learning (DL) models like artificial neural networks and deep neural networks were also reported for the identification of gene biomarkers[15]. Studies have used a deep neural network for classification between normal and tumor subjects with an accuracy of 97.9%[15]. The DL models capture the complex nonlinear relationships and features from the heterogeneous biological datasets. In this study, we have used recurrent neural networks (RNN), long short-term memory (LSTM), and one-dimensional convolutional neural networks (1D-CNN).

Drug repurposing is the strategy of utilizing existing medications to treat diseases other than those initially developed for cure[16]. Drug target interaction (DTI) describes the affinity of the drug to bind to a target, where lower  $K_d$  values indicate stronger and more effective binding[17]. DTIs determine a drug's mechanism of action, efficacy, and safety by defining binding ability to specific biological targets[18]. It can be analysed through the experimental and the computational approach. The experimental approach requires sophisticated equipment, cost, and infrastructure[19]. On the other hand, a computational approach can screen a large number of DTI and pharmacokinetics with less cost, time, and labor investment[20]. In this study, we followed a computational approach for drug repurposing.

The computational approach of DTI involves the generation of descriptors and embeddings of targets and drugs to predict the binding affinity. The targets can be represented by multiple descriptors like sequence-based (amino acid compositions, k-mers), physicochemical (hydrophobicity, charge distribution), and structural (secondary/tertiary features). These target embeddings can capture biological motifs, long-range dependencies, and evolutionary information[21, 22]. Drugs are represented by descriptors like physiological (molecular weight, logP, H-bond), topological (connectivity indices), and structural fingerprints (Morgan, molecular access system, extended connectivity fingerprints)[23]. However, conventional drug descriptors are predefined and fail to capture complex molecular structure, context, and biological interactions. Studies proposed the drug descriptors through DL models such as BioMLP and BioCNN by inheriting the ResMLP block to extract drug and protein

features[24]. These models can capture chemical intuition through substructures, integrating 3D geometry and sequential, semantic and evolutionary information. In this study, the drugs were represented through DL models such as Mol2Vec, graph isomorphism network (GIN), and Unimol, and targets were represented through ProtVec, ProtBert, and evolutionary scale modeling (ESM).

Drug-target interactions (DTI) are typically evaluated using computational toolboxes such as AutoDockTools, GLIDE, and AutoDock Vina. However, these methods are highly time-consuming and labor-intensive. Furthermore, DTI prediction becomes particularly challenging for targets and drugs whose 3D structures are not available. Studies used the artificial intelligence-based approaches to overcome the limitations of traditional molecular docking methods and, by leveraging large-scale data mining, uncover hidden biological patterns, thereby significantly enhancing prediction efficiency [25]. ML models such as Random Forest, support vector machine, Restricted Boltzmann Machines, and ensemble learning were proposed to improve DTI prediction[26]. Traditional ML models remain constrained by their dependence on handcrafted features and limited ability to capture complex biological relationships[27]. On the other hand, DL improves DTI prediction by automatically learning complex features from raw biological data, unlike traditional ML that relies on manual feature engineering. Studies have employed a multi-dimensional fusion DL model (MDF-DTA) integrating sequence, structure, and interaction features for drug-target binding affinity prediction, achieving superior accuracy and robustness[28]. A study has developed TeM-DTBA, a multimodal DL framework combining Mol2Vec, graph, and substructure fingerprints with ProtTrans and PSSM descriptors, using Lasso feature selection and CNN-based latent representation learning to predict drug-target binding affinity[29]. In this study, we employed DL techniques such as LSTM, RNN, and 1D-CNN for DTI prediction for PDAC.

Pharmacokinetics refers to the study of drug dynamics within the body across time, encompassing key ADMET properties such as absorption, distribution, metabolism, excretion, and toxicity[30]. It helps determine the optimal dose and ensures drug safety and efficacy. The conventional method for pharmacokinetic prediction of drugs can be evaluated through *in vivo* and *in vitro* methods. However, these techniques are often time-consuming, expensive, and resource-intensive[31]. Studies have employed models such as EdgeConv, graph convolutional networks [32], for ADMET property prediction. In this study, we employed DL techniques

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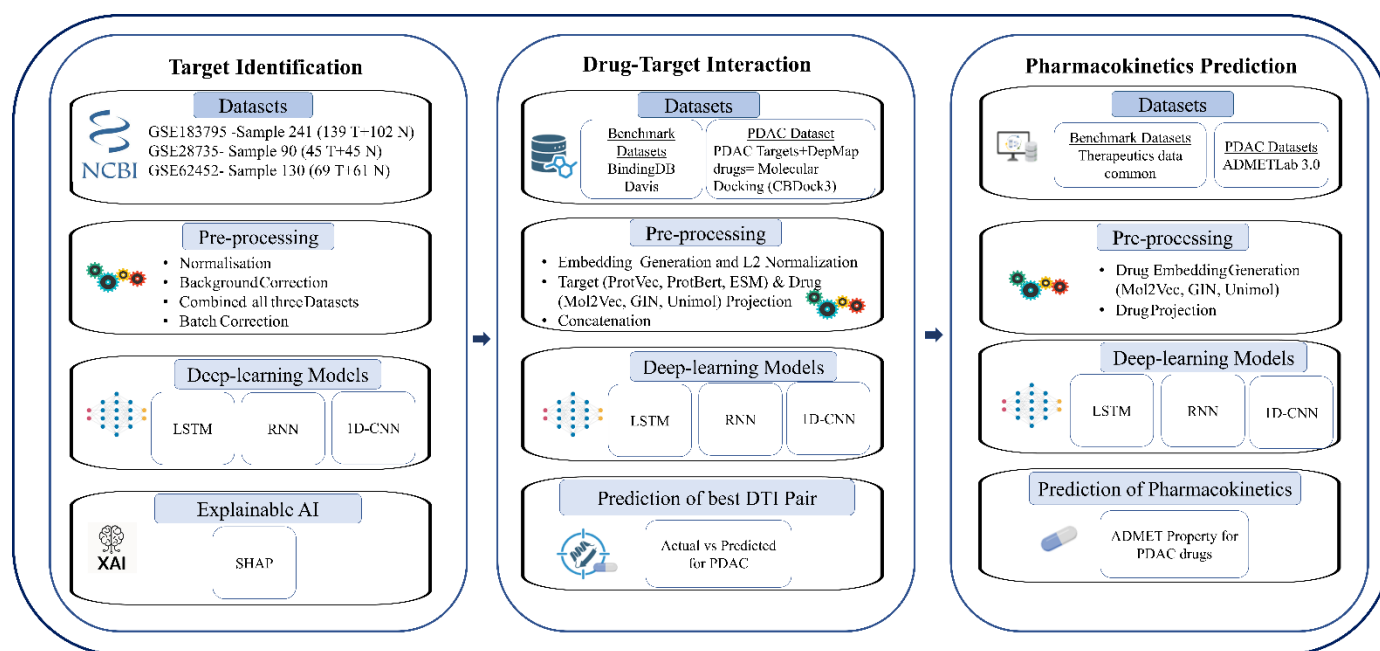


Fig. 1. Process pipeline of the study.

such as LSTM, RNN, and 1D-CNN for pharmacokinetics prediction in PDAC.

The key contributions of this study are as follows

- We proposed a novel framework for the identification of gene biomarkers for PDAC using LSTM, RNN, and 1D-CNN with SHapley Additive exPlanations (SHAP), combining three microarray gene expression datasets GSE183795, GSE28375, and GSE62452.
- We prepared the library of drugs using DepMap for PDAC, extracted the embeddings for targets using ProtVec, ProtBert, ESM, and drugs using Mol2Vec, GIN, Unimol, and employed LSTM, RNN, and 1D-CNN for the prediction of binding affinity for DTI.
- We predicted the pharmacokinetics of the drug for the PDAC library prepared using LSTM, RNN, and 1D-CNN with the same embeddings, Mol2Vec, GIN, and Unimol.

## II. MATERIALS AND METHODS

Figure 1 depicts the proposed process pipeline of the study. It involves (1) Identification of targets for PDAC, (2) DTI for the identified targets, and (3) Analyzing the pharmacokinetic properties of the drugs.

### 1. Targets Identification

#### A. Gene Expression dataset

In this study, we considered three microarray gene expression datasets from the National Center for Biotechnology Information, Gene Expression Omnibus (GEO), such as GSE183795, GSE28375, and GSE62452. These datasets

consist of 241 (PDAC=139, Adjacent non-tumour=102), 90 (PDAC=45, Adjacent non-tumour=45), and 130 (PDAC=69, Adjacent non-tumour=61) samples, respectively. The corresponding genes in each dataset were 19,245, 22,029, and 22,196. Initially, the raw gene expression data were retrieved from the GEO database using the GEOquery R package. The datasets were downloaded using the getGEO() function with the parameters GSEMatrix = TRUE to obtain the expression matrix and AnnotGPL = TRUE to include platform annotation information.

#### B. Pre-processing

The dataset was subjected to background correction using the normexp method implemented in the limma package, which adjusts for background signal and improves the accuracy of low-intensity measurements. Following this, quantile normalization was applied using the normalizeBetweenArrays function to standardize the distribution of expression values across all samples, thereby minimizing technical variability between arrays. We identified the common genes across the datasets and resulted with 19,042 genes. Batch effect correction was then performed on the integrated gene expression dataset using the ComBat algorithm implemented in the surrogate variable analysis in the R package. Samples were annotated with disease status (Normal and PDAC) and batch information (three GSE datasets). A design matrix incorporating disease status as a covariate was constructed to preserve biological variation associated with disease while correcting for technical variability. ComBat was then applied using an empirical Bayes framework to adjust for batch-associated variations. We evaluated the performance of the batch correction using principal component analysis before and after batch correction.

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The corrected expression matrix was subsequently used for further study[33].

### C. Deep learning models for target identification

We performed the ANOVA F-test to reduce the dimensionality and retained biologically informative features. The top  $K$  features were selected from the training data, with multiple values evaluated ( $K=20, 50, 100, 500, 1000$ ). The learned feature subset was applied to the independent test dataset without refitting, ensuring strict separation between training and evaluation data. The selected feature vector of size  $k$  was reshaped into a 2D sequence (seq len, input size), where:  $\text{seq len} \times \text{input size} = k$  and fed to three deep learning models, such as LSTM, RNN, and 1D-CNN. LSTM captures long-term dependencies in sequential data. RNN processes sequential data by retraining on previous information. 1D-CNN extracts local patterns and features from 1D sequential data [34-36]. Model performance was evaluated using stratified leave-one-out cross-validation, ensuring preservation of class distribution across folds. The model was assessed using multiple performance metrics such as accuracy, precision, sensitivity, specificity, F1-score, and area under the receiver operating characteristic curve (AUC). All analyses were conducted in Python using PyTorch for model development and scikit-learn for feature selection. Random seeds were fixed to ensure reproducibility across runs. Computations were performed on the GPU.

**Table 1** Parameters of deep learning models for target identification

| Parameters          | LSTM               | RNN                | 1D-CNN             |
|---------------------|--------------------|--------------------|--------------------|
| Layers              | 2                  | 2                  | 2                  |
| Hidden units        | 64                 | 64                 | 32                 |
| Dropout             | 0.3                | 0.3                | 0.3                |
| Activation Function | Sigmoid            | Sigmoid            | ReLU               |
| Optimizer           | AdamW              | AdamW              | AdamW              |
| Learning Rate       | $2 \times 10^{-4}$ | $1 \times 10^{-3}$ | $1 \times 10^{-3}$ |
| Weight Decay        | $10^{-4}$          | $10^{-5}$          | $10^{-5}$          |
| Epochs              | 30                 | 50                 | 50                 |

### D. Explainable artificial intelligence for target identification

We assessed the model's interpretability using SHAP with the DeepExplainer framework. A subset of training data was used as background input, and SHAP values were computed for the full test dataset. The feature importance was quantified as the mean absolute SHAP value across samples. The top-ranking features were visualized for each value of  $K$  to identify consistently informative genes[37].

### E. Survival analysis

We performed survival analysis on the identified targets using the gene expression profiling interactive analysis (GEPIA). It is

an online platform that enables interactive analysis of gene expression data derived from the cancer genome atlas (TCGA) and the genotype-tissue expression (GTEx) databases. It facilitates the exploration of gene expression patterns across various cancer types and normal tissues, along with their association with clinical outcomes. The survival plots module was utilized by entering the gene of interest, selecting overall survival, applying the group cut-off option, and specifying pancreatic cancer as the dataset. Statistical significance was determined based on  $p < 0.05$ [13].

## 2. Drug-target interaction prediction

### A. Drug library preparation and DTI datasets

We used the DepMap database to prepare the library of drugs for the identified targets for PDAC (22 targets). The top 15 drugs were considered for each target based on the correlation value. We received a total of 330 drugs (22 targets\*15 drugs), out of which a few were common across the targets, and ended up with 106 drugs (Detailed list is given in supplementary file -methods ). We obtained the binding energy for all possible 2,332 DTI pairs (22\*106) using CBDock3, which is used for training the model. We used two benchmark datasets, such as BindingDB and DAVIS[38], to compare the efficiency of the DL models. The BindingDB and DAVIS are publicly available dataset that contains information about the binding affinities of proteins and small molecules. The BindingDB dataset consists of 10,665 drugs and 1,413 targets (52,284 DTI pairs). The DAVIS dataset consists of 68 drugs and 379 targets (25,772 DTI pairs).

### B. Embeddings generation

We generated the embeddings for the targets (ProtVec, ProtBert, ESM) and drugs (Mol2Vec, GIN, Unimol). ProtVec (sequence-based embeddings) is a distributed representation of protein sequences learned from overlapping 3-mers using the skip-gram model. It encodes biological sequences into 1D-embeddings that capture functional and structural properties[21]. ProtBERT (Transformer-based embeddings) captures information about protein sequences based on the contacts between amino acids in their 2D structure. It is a bidirectional transformer language model trained on large protein sequence databases, adapting the BERT architecture to treat amino acids as tokens and entire protein sequences as sentences[22]. ESM is a family of large-scale protein language models. These models are trained on millions of protein sequences using transformer architectures, learning to predict masked amino acids in a sequence. It captures deep evolutionary patterns and structural information, enabling them to generate embeddings that represent protein structure, and function[39]. Mol2vec (molecular embeddings)[40] employs unsupervised learning on unlabelled datasets to generate embeddings for molecular substructures, which can be combined to represent entire compounds. It captures the sequential features of the drug by generating 1D drug embeddings of length equal to 300. GIN is pre-trained on unlabeled drug graphs. It is used to transform 2D embeddings

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into 1600-length feature vectors representing graph-based properties [36]. Uni-mol (3D embeddings) utilizes spatial conformations to learn rich representations, enabling the extraction of embeddings that encode both geometric and physicochemical characteristics of drugs[41].

### C. Deep Learning for DTI

We implemented three regression-based DL models, such as LSTM, RNN, and 1D-CNN, to analyze the DTI of PDAC. The binding affinity for the drug and target pairs was predicted by providing  $D = [d_1, d_2, d_3, \dots, d_m]$  and  $T = [t_1, t_2, t_3, \dots, t_n]$ , where  $D$  is the drug space,  $m$  is the number of drugs,  $T$  is the target space, and  $n$  is the number of targets. We trained the model with binding affinity values  $Y = \{y_{ij}^N\}$ , where  $y_{ij}$  is the binding score between drug<sub>*i*</sub>-target<sub>*j*</sub> pair and  $N$  is the number of observed affinities. We analyzed 19 combinations of drugs and target embeddings with 1 drug and 1 target (9 combinations of embeddings), 2 drugs and 2 targets (9 combinations of embeddings), and 3 drugs and 3 target embeddings (1 combination of embeddings). We adopted the 5-fold cross-validation strategy to train and evaluate the model using metrics such as the coefficient of determination ( $R^2$ ), root mean square error (RMSE), mean square error (MSE), and mean absolute error (MAE).

**Table 2** Parameters of deep learning models for DTI.

| Parameters          | LSTM               | RNN                | 1D-CNN             |
|---------------------|--------------------|--------------------|--------------------|
| Layers              | 1                  | 1                  | 3                  |
| Hidden units        | 64                 | 64                 | 64/128/256         |
| Dropout             | 0.2                | 0.2                | 0.3                |
| Activation Function | ReLU + tanh        | ReLU + tanh        | ReLU               |
| Optimizer           | AdamW              | AdamW              | AdamW              |
| Learning Rate       | $1 \times 10^{-3}$ | $1 \times 10^{-3}$ | $1 \times 10^{-3}$ |
| Weight Decay        | $10^{-5}$          | $10^{-5}$          | $10^{-5}$          |
| Epochs              | 100                | 100                | 100                |

## 3. Pharmacokinetics

### A. Datasets for pharmacokinetics

We analysed pharmacokinetics properties of 106 drugs for the PDAC. We obtained the five ADMET endpoints such as Caco-2 permeability, VDss, CYP2C9 inhibition, half-life, and hERG for PDAC using ADMETlab3.0. These values were used to train the models. We also evaluated the performance of the models using therapeutics data commons (TDC). It is a widely used open-science Python library that facilitates AI-driven drug

discovery and repurposing by providing curated datasets and standardized. The dataset comprises drugs annotated with ADMET properties. We analyzed 910, 1,128, 12,091, 665, and 2,92,915 compounds for absorption, distribution, metabolism, extraction, and toxicity, respectively, in the TDC dataset.

### B. Embeddings and deep learning for pharmacokinetics

We employed the same embeddings of the drugs, such as Mol2Vec, GIN, and Unimol, to predict the pharmacokinetic properties. Further, we considered 7 combinations of the drugs, like 1 embedding (3 combinations), 2 embeddings (3 combinations), and 3 embeddings (1 combination). We implemented classification (Metabolism) and regression-based (Absorption, Distribution, Excretion, Toxicity) DL models, such as LSTM, RNN, and 1D-CNN, to analyze the pharmacokinetics of PDAC. The models were evaluated using a 5-fold cross-validation strategy and computed metrics such as the MAE for absorption, Spearman correlation (SC) for distribution, excretion, and toxicity, and area under the receiver operating curve (AUROC) for metabolism.

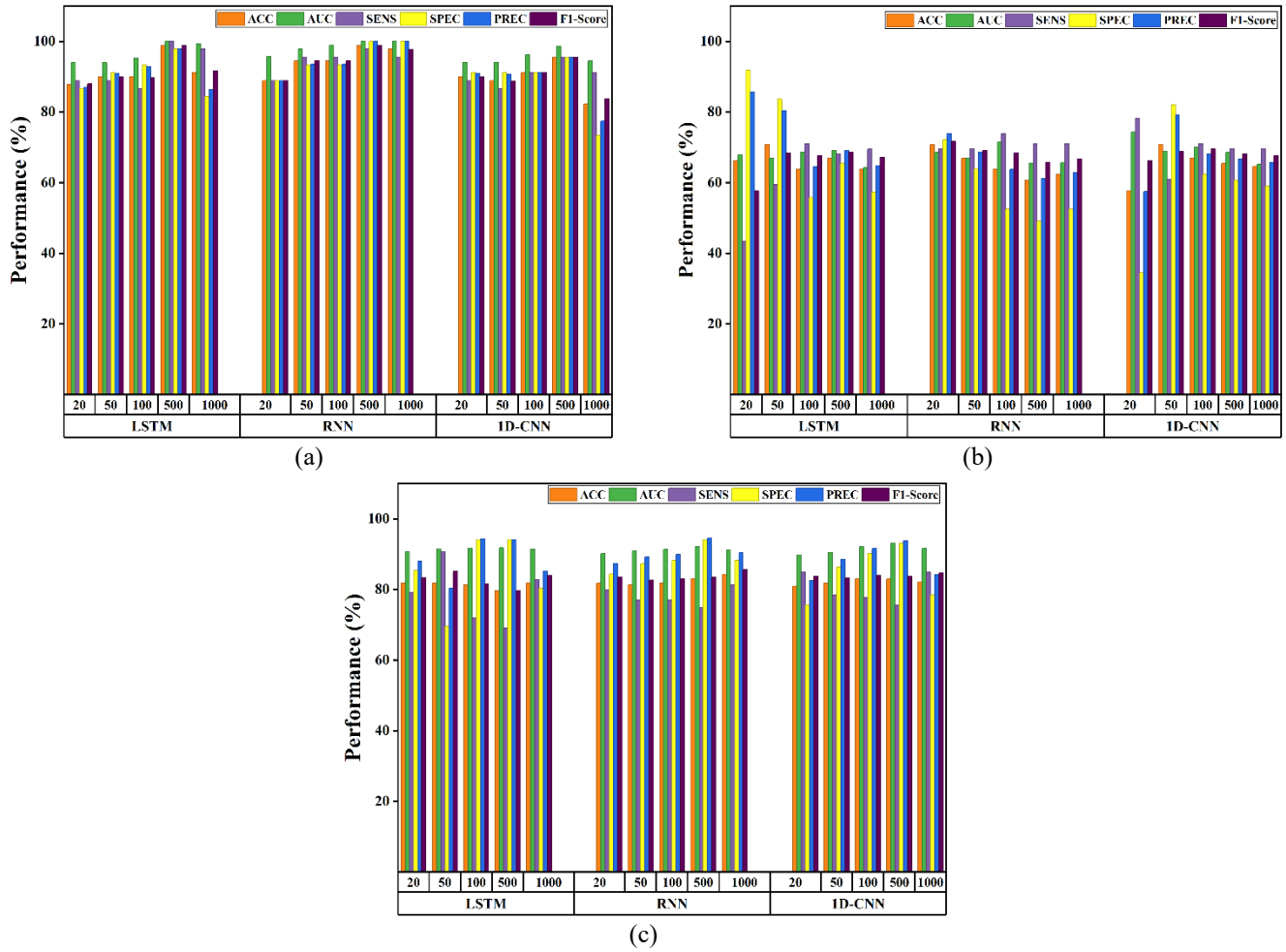
**Table 3** Parameters of deep learning models for Pharmacokinetics.

| Parameters          | LSTM               | RNN                | 1D-CNN             |
|---------------------|--------------------|--------------------|--------------------|
| Layers              | 1                  | 1                  | 3                  |
| Hidden units        | 64                 | 64                 | 64/128/256         |
| Dropout             | 0.2                | 0.2                | 0.3                |
| Activation Function | ReLU+ tanh         | ReLU+tanh          | ReLU               |
| Optimizer           | AdamW              | AdamW              | AdamW              |
| Learning Rate       | $1 \times 10^{-3}$ | $1 \times 10^{-3}$ | $1 \times 10^{-3}$ |
| Weight Decay        | $10^{-5}$          | $10^{-5}$          | $10^{-5}$          |
| Epochs              | 100                | 100                | 100                |

## III. RESULTS

Figure 2 (a) depicts the performance of the LSTM, RNN, and 1D-CNN models when training on combining two datasets, GSE62452 (130 samples), GSE183795 (241 samples), and testing on GSE28375 (90 samples). In each model, k=500 performed the best with LSTM, with an ACC, AUC, SENS, SPEC, PREC, and F1-Score of 98.89%, 100%, 100%, 97.78%, 97.83%, and 98.9%, respectively. We achieved similar performance on RNN with the corresponding performance metrics of 98.89%, 99.95%, 97.78%, 100%, 100%, and 98.88%. The performance dropped in 1D-CNN with the ACC,

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**Fig. 2.** Performance of LSTM, RNN, and 1D-CNN models trained on two datasets and tested on (a) GSE28375 (90 samples), (b) GSE62452 (130 samples), and (c) GSE183795 (241 samples). ACC-Accuracy, SENS-Sensitivity, SPEC-Specificity, PREC-Precision.

AUC, SENS, SPEC, PREC, and F1-Score of 95.56%, 98.67%, 95.56%, 95.56%, 95.56%, and 95.56%, respectively.

Figure 2 (b) shows the performance of DL models when training on combining two datasets, GSE183795 (241 samples), GSE28375 (90 samples), and testing on GSE62452 (130 samples). We achieved similar performance in all three models. We received ACC, AUC, SENS, SPEC, PREC, and F1-Score of 70.77%, 66.9%, 59.42%, 83.61%, 80.39%, and 68.33%, respectively, using the LSTM model (k=50). The RNN with k=20 produced corresponding metrics values of 70.77%, 68.54%, 69.57%, 72.13%, 73.85%, and 71.64%. Similarly, 1D-CNN at k=50, displayed 70.77%, 68.83%, 60.87%, 81.97%, 79.25%, and 68.85%. It can be noted that performance decayed with the GSE62452 (130 samples) as the test set.

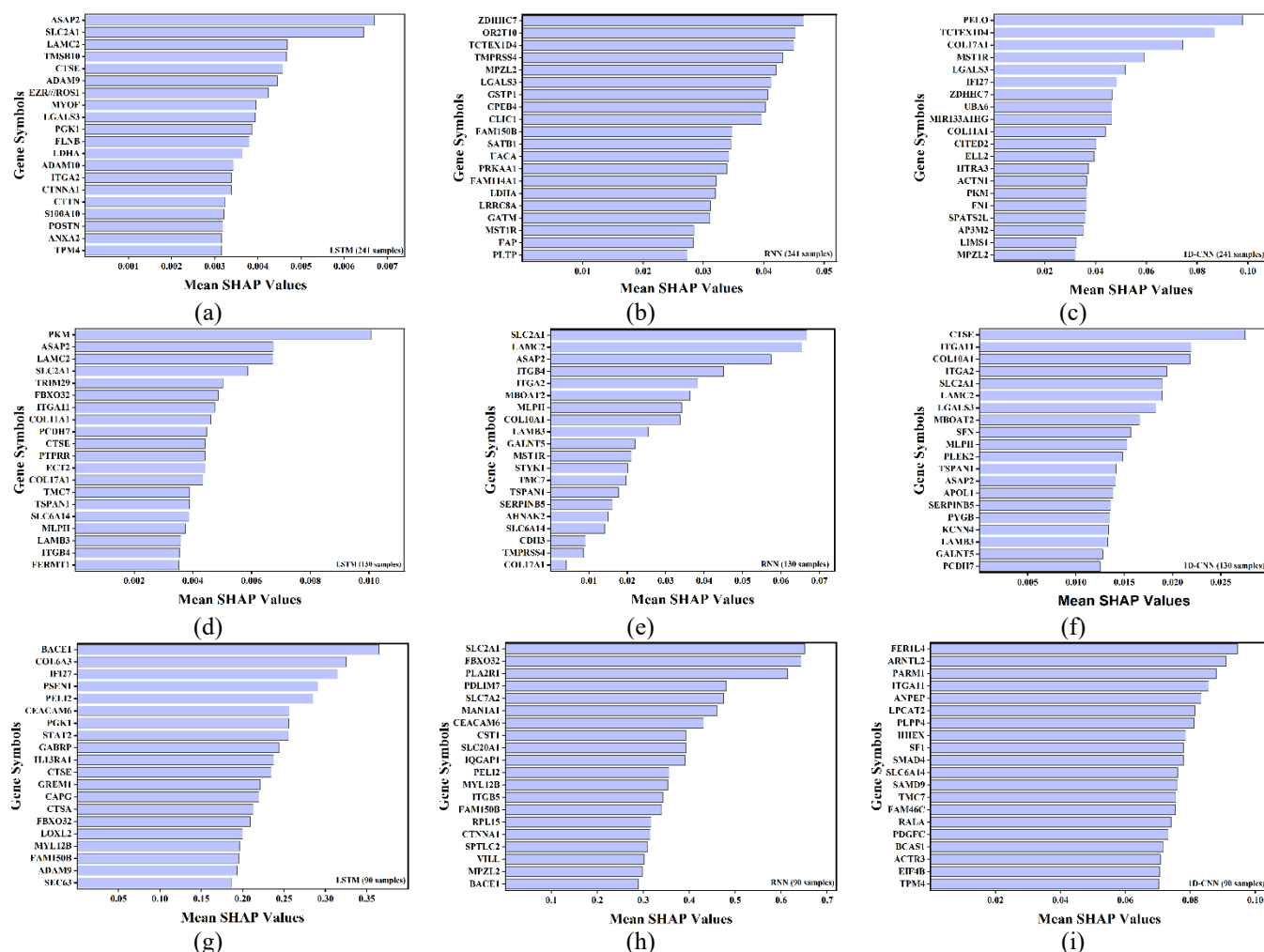
Figure 2 (c) represents the performance of the LSTM, RNN, and 1D-CNN models when trained on combining GSE62452 (130 samples), GSE28375 (90 samples), and tested on GSE183795 (241 samples). It can be noted that the performance improved in all the models. We found that LSTM with k=50 was able to produce the highest performance with the ACC, AUC, SENS, SPEC, PREC, and F1-Score of 81.74%,

91.5%, 90.65%, 69.61%, 80.25%, and 85.14%, respectively. It can be noted that the RNN required a large number of features (k=1000) to produce the peak performance of 84.23%, 91.1%, 81.29%, 88.24%, 90.4%, and 85.61%. The performance dropped in 1D-CNN with k=500 and produced evaluation metrics values of 82.99%, 93.13%, 75.54%, 93.14%, 93.75%, and 83.67%.

Figure 3 (a-i) depicts the top 20 features in each model using mean SHAP importance. In this plot, features are ranked from most to least important. The horizontal axis shows the mean absolute SHAP values, while the vertical axis depicts the gene symbols, providing a quantitative role of each feature (gene) contribution to the model's predictions. We found that 38 genes were common across all the models and datasets. It includes genes such as LDHA, ADAM9, COL17A1, MPZL2, ITGA2, LAMB3, SLC2A1, SERPINB5, TPM4, SLC6A14, TMC7, PKM, IFI27, TMRSS4, ASAP2, CTNNA1, TSPAN1, MST1R, GALNT5, MYL12B, LGALS3, FBXO32, COL11A1, TCTEX1D4, ITGB4, ITGA11, LAMC2, CEACAM6, PCDH7, COL10A1, ZDHHC7, PGK1, CTSE, MLPH, PELI2, MBOAT2, BACE1 and FAM150B. Our survival analysis



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**Fig. 3.** Summary of the important SHAP features when models were tested on (a-c) LSTM with GSE183795 (241 samples), RNN with GSE183795 (241 samples), 1D-CNN with GSE183795 (241 samples), (d-f) LSTM with GSE62452 (130 samples), RNN with GSE62452 (130 samples), 1D-CNN with GSE62452 (130 samples), (g-i) LSTM with GSE28375 (90 samples), RNN with GSE28375 (90 samples), 1D-CNN with GSE28375 (90 samples).

revealed 22 genes as statistically significant with  $p < 0.05$ . It includes LDHA ( $p < 0.0002$ ), ADAM9 ( $p < 0.00031$ ), COL17A1 ( $p < 0.00039$ ), MPZL2 ( $p < 0.00091$ ), ITGA2 ( $p < 0.0015$ ), LAMB3 ( $p < 0.0022$ ), SLC2A1 ( $p < 0.0043$ ), SERPINB5 ( $p < 0.0074$ ), TPM4 ( $p < 0.0079$ ), SLC6A14 ( $p < 0.0081$ ), TMC7 ( $p < 0.0088$ ), PKM ( $p < 0.0097$ ), IFI27 ( $p < 0.013$ ), TM6SF1 ( $p < 0.014$ ), ASAP2 ( $p < 0.016$ ), CTNNA1 ( $p < 0.016$ ), TSPAN1 ( $p < 0.023$ ), MST1R ( $p < 0.029$ ), GALNT5 ( $p < 0.035$ ), MYL12B ( $p < 0.04$ ), LGALS3 ( $p < 0.043$ ) and FBXO32 ( $p < 0.048$ ). These genes were considered for further analysis.

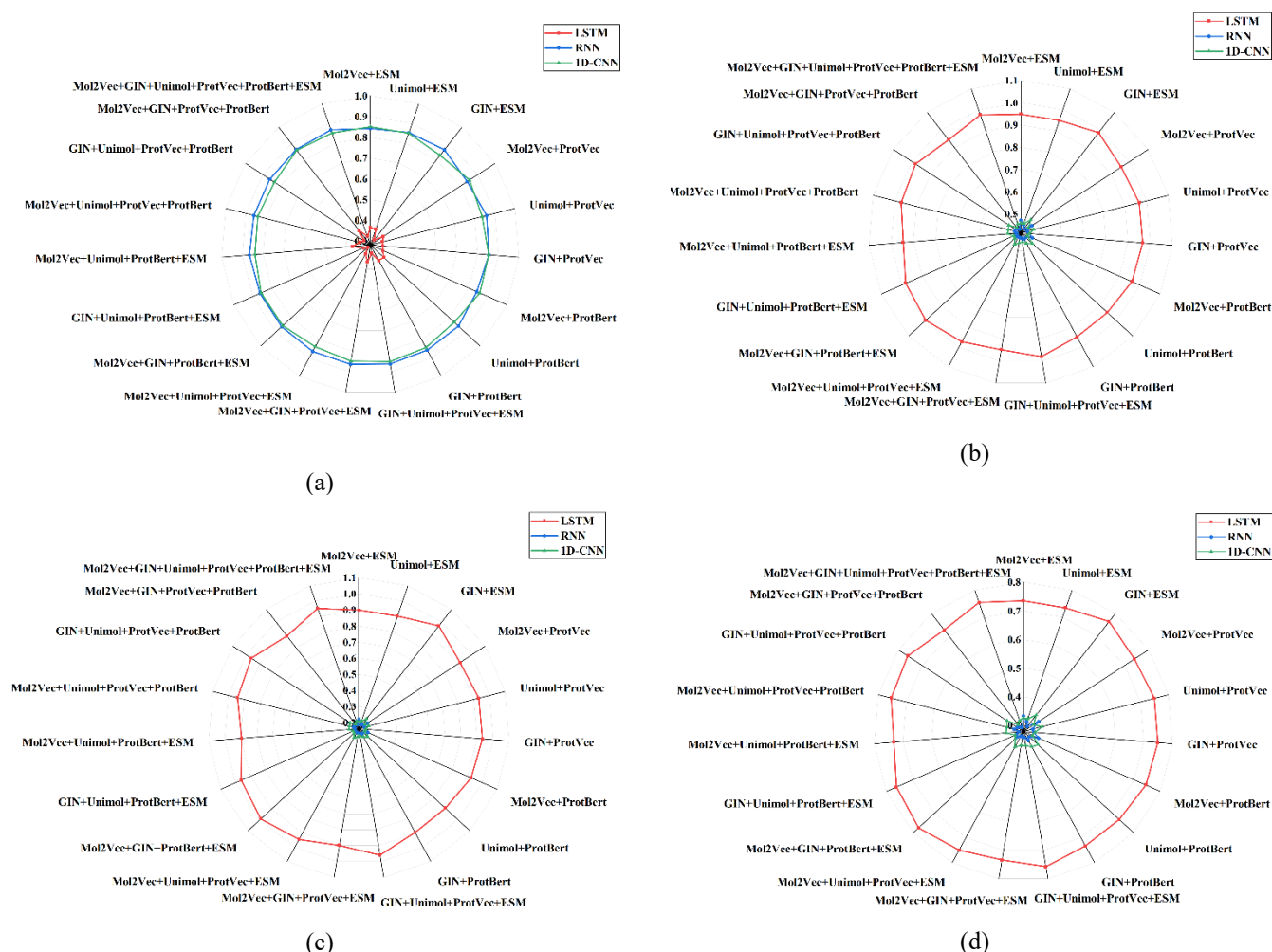
Figure 4 (a-d) represents the performance of the DL models on the DTIs PDAC dataset using the different combinations of the embeddings. In the case of LSTM, the combination Mol2Vec+GIN+ProtVec+ProtBert produced the  $R^2$ , RMSE, MSE, and MAE of 0.36, 0.94, 0.89, and 0.72, respectively. We received the highest performance on RNN using the Mol2Vec+GIN+Unimol+ProtVec+ProtBert+ESM combination with corresponding values of 0.86, 0.43, 0.18, and 0.29. Further on 1D-CNN, the combination of Mol2Vec+GIN+ProtBert+ProtVec produced the performance of 0.85, 0.44, 0.19, and 0.3. It can be noted that RNN performed

better than LSTM and 1D-CNN. We see that the Mol2Vec, GIN, ProtVec and ProtBert contributed well to all three models.

Supplementary Figure 1 (a-d) shows the performance of the DL models on the BindingDB dataset using the different combinations of the embeddings. We received the  $R^2$ , RMSE, MSE, and MAE of 0.44, 0.95, 0.9, and 0.66, respectively using the LSTM and the combination Mol2Vec+Unimol+ProtVec+ProtBert. The performance improved with RNN and GIN+ESM combination with the corresponding metrics values of 0.72, 0.67, 0.45, and 0.4. Performance dropped in the case of 1D-CNN with the combination of GIN+Unimol+ProtBert+ESM producing metrics values of 0.49, 0.91, 0.82, and 0.63. We can observe that RNN outperforms the other models on BindingDB dataset. LSTM and 1D-CNN perform well with the two drug and target embeddings, whereas RNN performs well on the one drug and target combination. It can be noted that Unimol, ProtBert, GIN and ESM consistently contributed well to at least two models.

Supplementary Figure 4 (e-h) depicts the performance of the DL models on the DAVIS dataset using the different combinations of the embeddings. The LSTM model with the

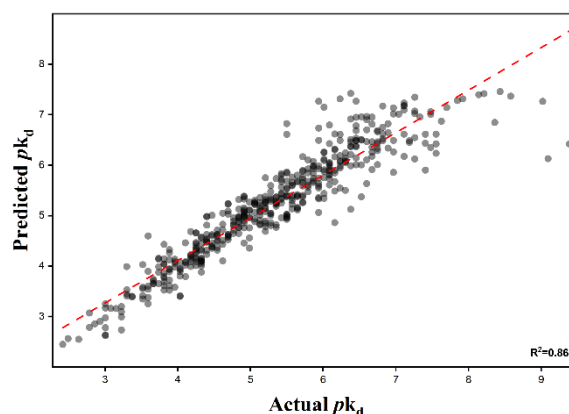
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**Fig. 4.** Performance of DL models on PDAC (a)  $R^2$  value, (b) RMSE, (c) MSE, and (d) MAE,

combination Mol2Vec+GIN+ProtBert+ESM produced the  $R^2$ , RMSE, MSE, and MAE of 0.09, 0.83, 0.7, and 0.56, respectively. The performance improved with the RNN and the GIN+Unimol+ProtVec+ProtBert combination, with corresponding metrics values of 0.61, 0.54, 0.29, and 0.29, respectively. The performance was reduced to 0.39, 0.68, 0.46, and 0.42 on 1D-CNN and the combination of Mol2Vec+GIN+ProtBert+ESM. Similar to earlier cases, we can observe that RNN outperforms LSTM and 1D-CNN. We can also see that the LSTM, RNN, and 1D-CNN performed well with the two drug and target embedding combinations. GIN and ProtBert were common in all three models. Further, Mol2Vec and ESM also contributed to at least two models.

Figure 5 depicts the relation between actual vs predicted binding affinity for the best performing model and embeddings (RNN and Mol2Vec+GIN+Unimol+ProtVec+ProtBert+ESM) on the PDAC dataset. It can be noted that the model produced an  $R^2$  value of 0.62. We can observe that most of the fits lie near the linear fit with minimum outliers. This shows the better performance of the model with minimal error.



**Fig. 5.** Actual vs predicted binding affinity ( $pK_d$ ) for the best-performing model for the PDAC dataset.

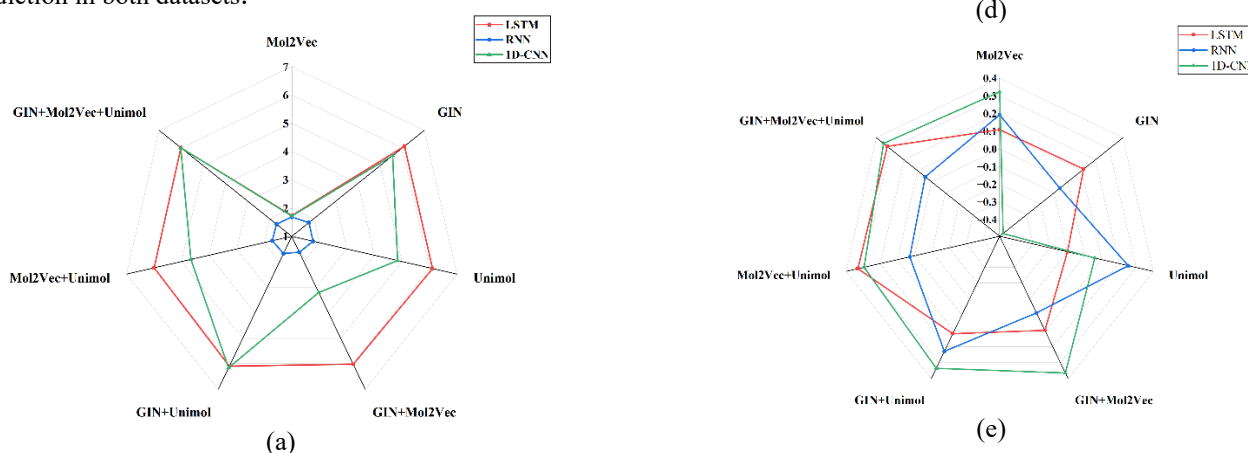
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Figure 6 (a) and supplementary Fig.2 (a) depict the MAE for the absorption endpoint for PDAC and TDC datasets, respectively, using various combinations of embeddings. Our pipeline involving RNN and GIN+Mol2Vec embeddings produced a MAE of 1.62. In the case of the TDC dataset, we achieved the highest performance of 0.51 with RNN and GIN+Mol2Vec. It can be noted that RNN and GIN+Mol2Vec performed consistently well across both datasets.

Figure 6 (b) and supplementary Fig.2 (b) shows the SC for the distribution endpoint for PDAC and TDC datasets. We received the highest performance using LSTM and Mol2Vec embeddings with SC of 0.56. It can be noted that the TDC dataset produced lower results with an SC of 0.06 using 1D-CNN and Unimol. We can see that the distribution parameters of the PDAC dataset are relatively better than the TDC. It may be due to the greater variability in the data distribution. Figures 6 (c) and supplementary Fig.2 (c) represent the AUROC metrics for the metabolism endpoint for PDAC and TDC datasets. We can see that the 1D-CNN and Mol2Vec+Unimol embeddings produced an AUROC of 62.5%. On the other hand, we achieved a highest performance of 82.57% using RNN and GIN+Mol2Vec on the TDC dataset. It can be noted that Mol2Vec performed well in both datasets, depicting the ability to capture metabolic characteristics.

Figure 6 (d) and supplementary Fig.2 (d) illustrate the SC performance metrics for the excretion endpoint for PDAC and TDC datasets. Our models produced reasonable performance with SC of 0.32 using LSTM and Unimol embeddings on the PDAC dataset. It can be noted that performance improved on the TDC dataset with SC of 0.36 using LSTM and GIN. It can be noted that LSTM produced well for the excretion parameters, with similar performance for both the datasets.

Figure 6 (e) and supplementary Fig.2 (e) shows the SC metrics for the toxicity endpoint for PDAC and TDC datasets. We received SC of 0.36 using 1D-CNN and GIN+Mol2Vec embeddings on the PDAC dataset. The performance was highly improved on the TDC dataset with SC of 0.7 using 1D-CNN and GIN+Mol2Vec+Unimol. We can see that the 1D-CNN, GIN, and Mol2Vec consistently contributed well to the prediction in both datasets.



**Fig. 6.** Performance of DL models on PDAC dataset (a) MAE (Absorption), (b) SC (Distribution), (c) AUROC (Metabolism), (d) SC (Excretion), (e) SC (Toxicity)

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#### IV. DISCUSSION

In this study, we used three microarray gene expression datasets (GSE183795, GSE28375, and GSE62452) to identify a robust PDAC-causing genetic biomarker for prognosis and therapy prediction. Initially, we identified DEGs between PDAC and normal samples using DL approaches with training on two datasets at a time and testing on the third dataset. We performed survival analysis and enlisted 22 DEGs for the PDAC. Further, we prepared a library of drugs and carried out DTI and pharmacokinetics analysis.

##### A. Effect of deep learning algorithm on biomarker identification

We implemented LSTM, RNN, and 1D-CNN for target identification and classification in PDAC. Our results revealed that the performance trends are largely driven by variations in training data size and model complexity. LSTM and RNN achieved optimal performance (accuracy of 98.89%) when trained on larger datasets (241 and 130 samples), highlighting their ability to capture complex patterns. A deterioration in performance with reduced training data and larger test sets indicates limited exposure to data variability, leading to poorer generalization. It leads to a tendency to favor the majority class, as reflected by imbalanced sensitivity and specificity. A study has used a large number of samples from the physionet and tested on a small number of samples from other datasets and achieved higher results compared to vice versa[42]. We received better performance on a smaller ( $k=20$  and  $50$ ) and larger number of features ( $k=500$  and  $1000$ ). It was inconsistent

between the models. We found that LSTM and RNN performed well compared to 1D-CNN in discriminating between normal and PDAC. It may be due to the capability of LSTM and RNN in capturing temporal patterns and sequential data. A study has compared the LSTM, RNN, 1D-CNN and found that LSTM performs better followed by 1D-CNN and RNN[43]. Studies have used artificial neural networks and deep neural networks to classify the normal and PDAC samples and reported lower accuracy than our models[44]. However, none of the previous studies used the LSTM and RNN for target detection in PDAC from the transcriptomic data

##### B. Effects of DEGs

We found 22 genes as the biomarker for PDAC (LDHA, ADAM9, COL17A1, MPZL2, ITGA2, LAMB3, SLC2A1, SERPINB5, TPM4, SLC6A14, TMC7, PKM, IFI27, TMRSS4, ASAP2, CTNNA1, TSPAN1, MST1R, GALNT5, MYL12B, LGALS3, and FBXO32). Further, we reviewed existing literature describing the biological roles of these genes. We summarised these gene pathophysiological pathways underlying PDAC development in Table 4.

##### C. Effect of drug and target embeddings on DTI Prediction

Our study indicates that the Mol2vec and GIN embeddings consistently contributed to model performance across all datasets and models. In addition, protein embeddings ProtBert and ProtVec frequently occur in top-performing combinations.

**Table 4** Studies reported genes identified in our analysis for PDAC

| Study | Gene Name | Condition         | Role                                                                                               |
|-------|-----------|-------------------|----------------------------------------------------------------------------------------------------|
| [45]  | LDHA      | Pancreatic cancer | Promotes aerobic glycolysis and cancer cell growth                                                 |
| [46]  | ADAM9     | Pancreatic cancer | EGFR/MAPK signaling pathway enhance cell adhesion and invasion via integrin interactions           |
| [47]  | COL17A1   | Pancreatic cancer | Tumor growth, migration, and invasion through epithelial-mesenchymal transition (EMT) pathway      |
| [48]  | MPZL2     | Pancreatic cancer | Promotes tumor progression by modulating cell adhesion, migration, and immune evasion              |
| [49]  | ITGA2     | Pancreatic cancer | Genomic stability                                                                                  |
| [50]  | LAMB3     | Pancreatic cancer | Promotes proliferation, invasion, migration, and metastasis through activation of the PI3K pathway |
| [51]  | SLC2A1    | Pancreatic cancer | Enhanced glycolysis and tumor aggressiveness under hypoxic conditions                              |
| [52]  | SERPINB5  | Pancreatic cancer | Invasion, metastasis, and EMT                                                                      |
| [53]  | TPM4      | Gastric cancer    | Cytoskeletal remodeling, migration, and invasion,                                                  |
| [54]  | SLC6A14   | Pancreatic cancer | Promotes glutamine uptake, immune evasion                                                          |
| [55]  | TMC7      | Pancreatic cancer | Liver metastasis                                                                                   |

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|      |         |                   |                                                                                   |
|------|---------|-------------------|-----------------------------------------------------------------------------------|
| [56] | PKM     | Pancreatic cancer | Promoting metabolism, migration                                                   |
| [57] | IFI27   | Pancreatic cancer | Glycolysis and promotes immune suppression                                        |
| [58] | TMPRSS4 | Pancreatic cancer | Proliferation, invasion, and immune evasion via ERK1/2 signaling pathway          |
| [59] | ASAP2   | Pancreatic cancer | Proliferation, invasion, and metastasis through oncogenic c-MET-related signaling |
| [60] | CTNNA1  | Pancreatic cancer | SETD2 loss and the hsa-miR-30d-5p/GJA1 axis to promote EMT                        |
| [61] | TSPAN1  | Pancreatic cancer | Proliferation, migration, invasion, and tumorigenesis                             |
| [62] | MST1R   | Pancreatic cancer | Metastasis through PI3K/AKT and MAPK signaling                                    |

It can be noted that Mol2Vec refined the embeddings of molecular substructures to minimize the prediction error. It effectively captured the relationships and contextual information encoded by the SMILES. The GIN model was able to capture the topological features and relationships between atoms and bonds based on the molecular graphs[28]. ProtBert provided crucial semantic information from short sequences later on longer sequences, improving the efficiency of the model[63]. ProtVec captured the contextual relationship between k-mers embedding based on the word2vec, which gives meaningful biological insights from the sequence. These embeddings were able to extract the parameters well on their own from the sequence and structure motifs directly, rather than depending on the traditional descriptors[64]. A recent study has reported that the Mol2vec and ESM contributed well to the DL based DTI model[65]. A study has fused Mol2vec, GIN, equivariant graph neural network and Provec, Probert, ESM for the DTI prediction using DL models[28]. However, studies never analysed the effect of different combinations of target and drug embeddings for the DTI prediction.

#### D. Effect of deep learning algorithm on DTI Prediction

The comparative analysis of the DL models across the BindingDB, DAVIS, and PDAC datasets demonstrated that RNN consistently achieved superior predictive performance compared to LSTM and 1D-CNN. Among the embedding combinations, hybrid representations of drug and protein embeddings significantly improved the DTI prediction. Studies have proposed DeepConv-DTI, a convolutional neural network model that extracts local residue patterns directly from raw protein sequences and combines them with drug fingerprints, outperforming descriptor based and prior DL models in large-scale DTI prediction[66]. Another study has introduced a DL framework that learns latent features directly from raw drug and protein data, outperforming traditional similarity-based and matrix factorization methods in large-scale DTI prediction[67].

#### E. Effect of embeddings and deep learning algorithm on pharmacokinetics prediction

We found that Mol2vec alone or along with GIN and Unimol embeddings contributed well to the model. Mol2vec captures chemical substructure in a continuous vector space which helps in better prediction of ADMET property[68]. The model performance varies with the ADMET parameters. Our results revealed that RNN performed well in absorption, LSTM in distribution and excretion and 1D-CNN in metabolism and toxicity. A study has proposed MolGIN based on GIN which captures molecular graph topology by aggregating atom-level neighborhood information for ADMET prediction[69]. Another study showed that descriptor-augmented Mol2Vec embeddings enhanced ADMET prediction accuracy, achieving lower absorption error and higher correlation/AUPRC scores across, distribution, metabolism, excretion, and toxicity compared with baseline Mol2Vec[68]. A study has compared SVM, 1D-CNN, and kernel partial least squares-CNN for the ADMET classification, in which a CNN-based architecture performed well[70]. However, studies have never compared the combinations of embeddings with DL for ADMET prediction.

## V. LIMITATIONS AND FUTURE SCOPE

We comprehensively established a pipeline for drug repurposing for PDAC from target identification, DTI and pharmacokinetics. The study has few limitations such as usage of microarray gene expression data for target identification rather than RNA-seq data. Although multiple embedding techniques (ProtVec, ProtBert, ESM, Mol2Vec, GIN, and Unimol) were used, the study primarily focused on sequence and structure-based embeddings. Other biological factors such as epigenetics, tumor microenvironment, protein conformational dynamics, and pathway-level interactions were not incorporated. Pharmacokinetic prediction tasks achieved only moderate performance metrics for the candidate drugs. In future, we will incorporate multiomics data, and the experimental validation using *in vitro* and *in vivo* studies. clinically relevant insights into drug repurposing for PDAC.

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## VI. CONCLUSIONS

Our study proposed a novel process pipeline for identifying prognostic biomarkers, predicting DTI, and evaluating pharmacokinetic properties in PDAC. We integrated multiple gene expression datasets, advanced embedding techniques and DL models. We identified 22 significant biomarker genes with a high balanced accuracy of 98.89% using the RNN model. Furthermore, the combined embedding strategy of Mol2Vec, UniMol, ProtVec, and ProtBert with RNN achieved strong DTI prediction performance ( $R^2 = 0.62$ ), leading to the identification of 106 potential therapeutic drugs for PDAC. In addition, our pharmacokinetic prediction models evaluated ADMET properties using different combinations of molecular embeddings and DL architectures, demonstrating reliable predictive capability across absorption, distribution, metabolism, excretion, and toxicity parameters. The superior performance of Mol2Vec and GIN-based representations highlights the importance of capturing both chemical semantics and molecular graph topology in drug discovery tasks. The proposed framework provides a scalable and efficient computational strategy for biomarker discovery, drug repurposing, and personalized therapeutic development in PDAC.

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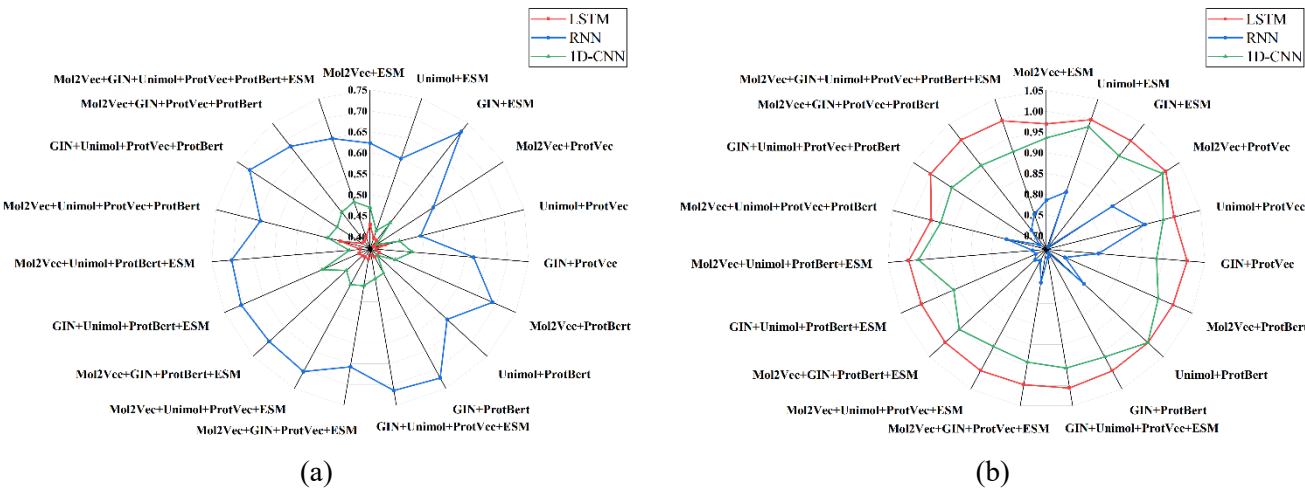


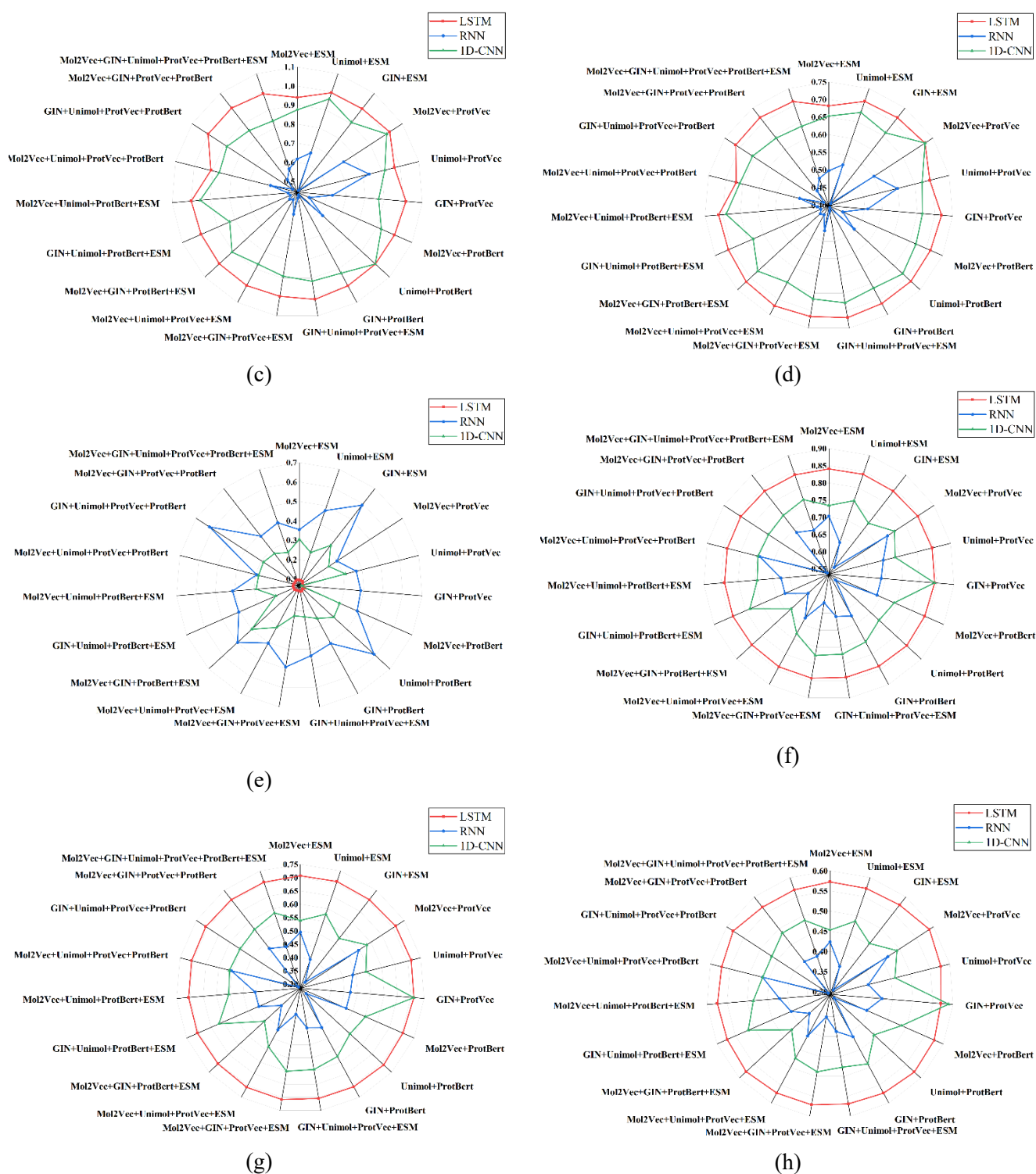
1Methods

2PDAC (In-house drug dataset)

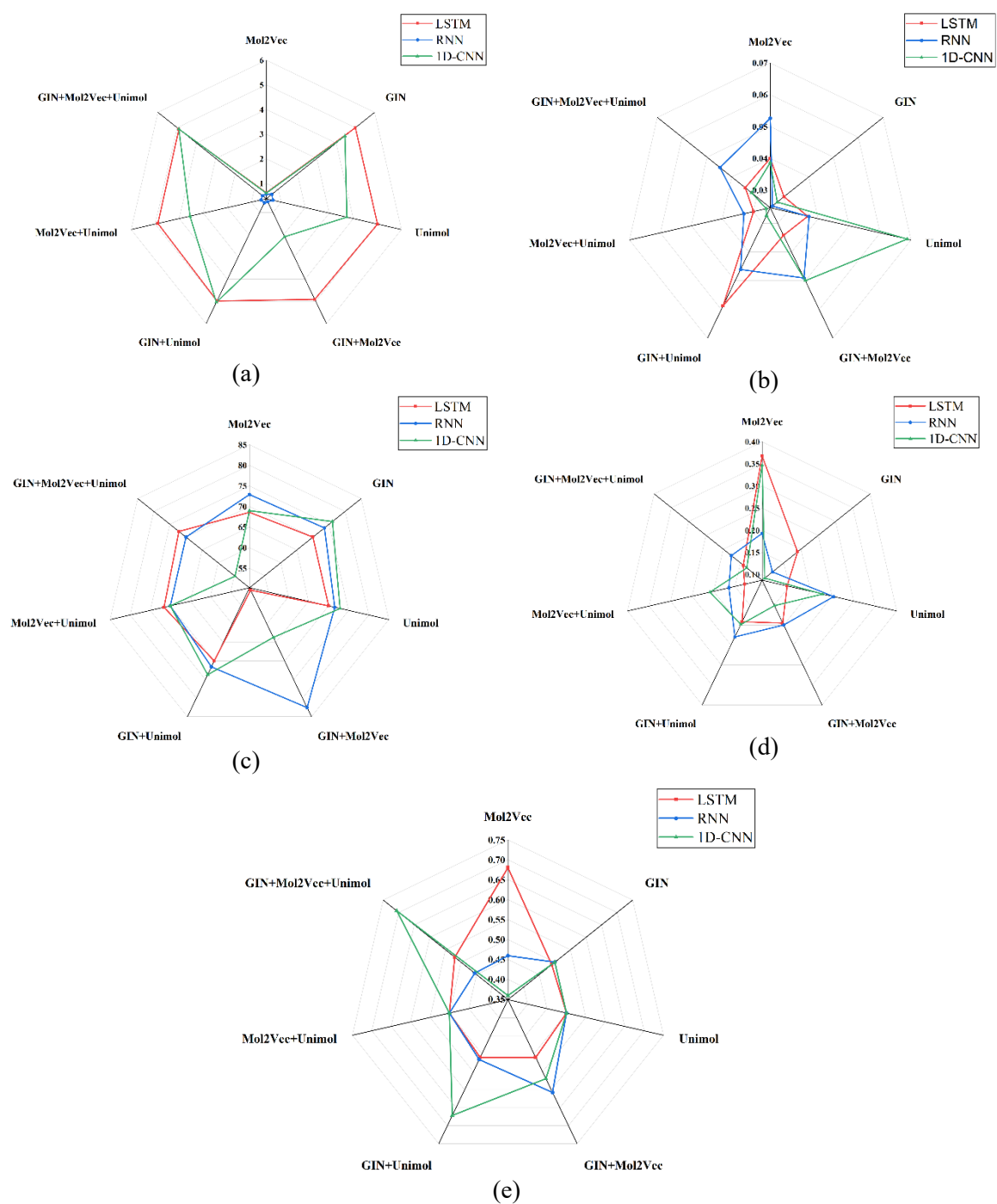
3 We continued with 106 drugs as considered for the DTI study. As we have 22 targets and considered  
4 15 drugs so that total drug list is 330, but some drugs are common between (-JQ-1 (2), AB1010 (2),  
5 ACY-1215(2), ANASTROZOLE (2), APO866(8), AT7867(3), BELINOSTAT (7), BI-2536(3), BI-  
6 97C1(8), BIRB 796 (4), BMS-754807(2), BRD9876:MK-1775 (2), BRD-A94377914 (5), BRD-  
7 K30748066 (6), CAPSAICIN(2), CARBOPLATIN:ETOPOSIDE (5), CCT036477 (5), CD532 (3),  
8 CHLORAMPHENICOL-PALMITATE (2), CICLOPIROX(2), CLOFARABINE(3), COMPOUND  
9 23 CITRATE(2), CYTARABINE HYDROCHLORIDE (8), DECITABINE (6),  
10 DECITABINE:CARBOPLATIN (3), DOXORUBICIN (5), ENCORAFENIB(3), ENTINOSTAT(2),  
11 EPIRUBICIN (2), ETOPOSIDE (8), GSK 269962A (2), GSK461364 (3), GSK-J4 (9), IMD-0354 (2),  
12 ISOMETHEPTENE-MUCATE (2), KB NB 142-70 (2), KPT185 (3), KW-2449 (3), KX2-391(2),  
13 LOXISTATIN-ACID (3), LRRK2-IN-1 (5), LY-2183240 (6), MERCK60(2), MITOXANTRONE (3),  
14 ML162(2), ML210(4), MOLIBRESIB(4), MST 312(6), NORETHINDRONE-ACETATE(5), NVP-  
15 BVU972(2), PF-3758309 (3),PHENELZINE(4), PRIMA-1 (5), PX-12(11), QW-BI-011 (3), SB-  
16 225002(3), TAME(3), TENIPOSIDE (14), THICOLCHICOSIDE (4), TIVANTINIB(5),  
17 TRIAZOLOTHIADIAZINE(5), UH-232-(+)(2), VELBAN(2), VINCRISTINE(8),  
18 VORINOSTAT(10), VX-680(5). In addition, for few drugs no SMILES was available such as 1S,3R-  
19 RSL-3, BI-97C1, BRD9876:MK-1775, CARBOPLATIN:ETOPOSIDE, CHLORAMPHENICOL-  
20 PALMITATE, DECITABINE: CARBOPLATIN, NSC48300, QW-BI-011,  
21 TANESPIMYCIN:GEMCITABINE, TIPIFARNIB-P2, VELBAN, VENETOCLAX, VINCRISTINE,  
22 VORINOSTAT:CARBOPLATIN.

24Results





**Supplementary Figure 1.** Performance of DL models on BindingDB (a) R<sup>2</sup> value, (b) RMSE, (c) MSE, (d) MAE, DAVIS (e) R<sup>2</sup> value, (f) RMSE, (g) MSE, (h) MAE



**Supplementary Figure 2.** Performance of DL models on TDC Dataset (a) MAE (Absorption), (b) SC (Distribution), (c) AUROC (Metabolism), (d) SC (Excretion), and (e) SC (Toxicity).